



STARBOOKS

Whiz Challenge

AY 2025-2026

AUTHORS:

Kelly Dumbrigue
Arcielle Marie Gercan
Shandrae Lois Quianzon
Janice Maxene Salipande
NEXUS

CONTRIBUTES TO:

4 QUALITY
EDUCATION



9 INDUSTRY, INNOVATION
AND INFRASTRUCTURE



10 REDUCED
INEQUALITIES



INDIRECTLY AIDS:

8 DECENT WORK AND
ECONOMIC GROWTH



**Entity Relationship Diagram for
DOST STARBOOKS: Whiz Challenge**

Project Documentation Submitted to the Faculty of the
School of Computing and Information Technologies
Asia Pacific College

In Partial Fulfillment of the Requirements for
Applied Project for CS
MCSPROJ

Submitted by:

Dumbrique, Kelly (Team Leader)

Gercan, Arcielle Marie (Member)

Quianzon, Shandrae Lois (Member)

Salipande, Janice Maxene (Member)

NEXUS – SF 231

Submitted to:

Mr. Porfirio Atienza

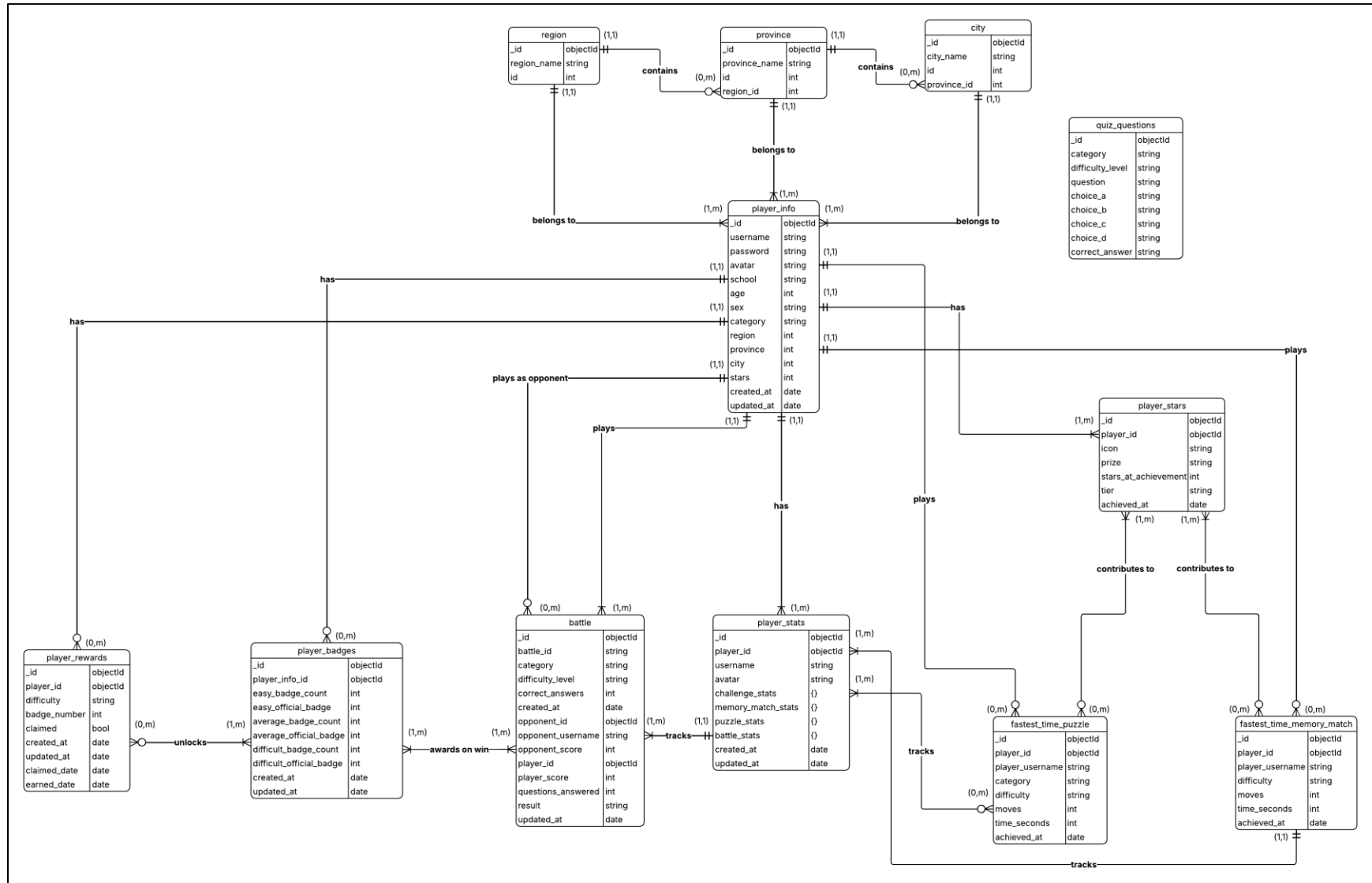
Course Instructor

January 2026

2nd Term

A.Y. 2025 – 2026

ENTITY RELATIONSHIP DIAGRAM



DATA DICTIONARY

ENTITY	ATTRIBUTE	CONTENTS	DATA TYPE	FORMAT	RANGE	REQUIRED	PK OR FK	FK REF TABLE
player_info	_id	Player ID	ObjectId	ObjectId	-	Y	PK	
	username	Player Username	String	Text	-	Y		
	password	Player Password	String	Text	-	Y		
	age	Player Age	INT	Number	-	Y		
	sex	Player Sex	String	Text	-	Y		
	avatar	Player Avatar Image	String	Text	-	Y		
	category	Player Category	String	Text	-	Y		
	region	Geographic Region ID	INT	Number	-	Y	FK	region
	province	Geographic Province ID	INT	Number	-	Y	FK	province
	city	Geographic City ID	INT	Number	-	Y	FK	city
	school	School Name	String	Text	-	Y		
	stars	Total Stars Accumulated	INT	Number	>=0	Y		
	created_at	Record Creation Stamp	DATE	ISO 8601	-	Y		
	updated_at	Record Last Update Stamp	DATE	ISO 8601	-	Y		
battle	_id	Battle ID	ObjectId	ObjectId	-	Y	PK	
	battle_id	Battle Session ID	String	Text	-	Y		

	player_id	Player ID	Objectld	Objectld	-	Y	FK	player_i nfo
	opponent_id	Opponent ID	Objectld	Objectld	-	Y	FK	player_i nfo
	opponent_use rname	Opponent Username	String	Text	-	Y		
	category	Battle Category (math/science)	String	Text	-	Y		
	difficulty_level	Battle Difficulty Level	String	Text	easy/ave rage/diffi cult	Y		
	player_score	Player Score	INT	Number	>=0	Y		
	opponent_sco re	Opponent Score	INT	Number	>=0	Y		
	questions_an swered	Number of Questions Answered	INT	Number	>=0	Y		
	correct_answ ers	Number of Correct Answers	INT	Number	>=0	Y		
	result	Battle Result	String	Text	win/loss/ draw	Y		
	created_at	Record Creation Stamp	DATE	ISO 8601	-	Y		
	updated_at	Record Last Update Stamp	DATE	ISO 8601	-	Y		
fastest_ti me_me mory_m atch	_id	Unique ID	Objectld	Objectld	-	Y	PK	
	player_id	Player ID	Objectld	Objectld	-	Y	FK	player_i nfo
	player_userna me	Player Userername	String	Text	-	Y		

	difficulty	Difficulty Level	String	Text	-	Y		
	time_seconds	Time Taken	INT	Number	>0	Y		
	moves	Number of moves made	Int	Number	>=0	Y		
	achieved_at	Achievment stamp	DATE	ISO 8601	-	Y		
fastest_time_puzzle	_id	Unique ID	ObjectID	Objectld	-	Y	PK	
	player_id	Player ID	ObjectID	Objectld	-	Y	FK	player_info
	player_username	Player Username	String	Text	-	Y		
	category	Puzzle Category (math/science	String	Text	-	Y		
	difficulty	Puzzle Difficulty Level	String	Text	easy/average/difficult	Y		
	time_seconds	Time Taken	INT	Number	>0	Y		
	moves	Number of moves made	Int	Number	>=0	Y		
	achieved_at	Achievment stamp	DATE	ISO 8601	-	Y		
player_badges	_id	Unique ID	ObjectID	Objectld	-	Y	PK	
	player_info_id	Player ID	ObjectID	Objectld	-	Y	FK	player_info
	easy_badge_count	Count of easy badges earned	INT	Number	>=0	Y		
	easy_official_badge	Official easy badge status	INT	Numer	-	Y		
	average_badge_count	Count of average	INT	Number	>=0	Y		

		badges earned						
	average_official_badge	Official average badge status	INT	Numer	-	Y		
	difficult_badge_count	Count of difficult badges earned	INT	Number	>=0	Y		
	difficult_official_badge	Official difficult badge status	INT	Numer	-	Y		
	created_at	Record Creation Stamp	DATE	ISO 8601	-	Y		
	updated_at	Record Last Update Stamp	DATE	ISO 8601	-	Y		
player_rewards	_id	Unique ID	ObjectID	ObjectId	-	Y	PK	
	player_id	Player ID	ObjectID	ObjectId	-	Y	FK	player_info
	fastest_time_record_id	Reference to fastest time record	ObjectID	ObjectId	-	Y	FK	fastest_time_memory_match / fastest_time_puzzle
	game_mode	Game mode for reward	String	Text	-	Y		
	difficulty	Difficulty Level	String	Text	-	Y		
	reward_type	Reward Type	String	Text	-	Y		

	reward_message	Reward description message	String	Text	-	Y		
	reward_stars	Stars Rewarded	INT	Number	>=0	Y		
	time_seconds	Time Taken	INT	Number	>0	Y		
	claimed	Whether reward is claimed	Boolean	true/false	-	Y		
	earned_date	Date reward was earned	DATE	ISO 8601	-	Y		
	claimed_date	Date reward was claimed	DATE	ISO 8601	-	Y		
	created_at	Record Creation Stamp	DATE	ISO 8601	-	Y		
	updated_at	Record Last Update Stamp	DATE	ISO 8601	-	Y		
player_stars	_id	Unique ID	ObjectID	ObjectId	-	Y	PK	
	player_id	Player ID	ObjectID	ObjectId	-	Y	FK	player_info
	tier	Star Tier Level	String	Text	-	Y		
	icon	Icon identifier for tier	String	Text	-	Y		
	prize	Prize Description	String	Text	-	Y		
	stars_at_achievement	Stars count at achievement	INT	Number	>=0	Y		
	achieved_at	Achievement Timestamp	DATE	ISO 8601	-	Y		
player_stats	_id	Unique ID	ObjectID	ObjectId	-	Y	PK	
	player_id	Player ID	ObjectID	ObjectId	-	Y	FK	player_info

	username	Player username	String	Text	-	Y		
	avatar	Player Avatar Image	String	Text	-	Y		
	battle_stats	Battle Statistics	Object	JSON	-	Y		
	battle_stats.math	Math battle stats	Object	JSON	-	Y		
	battle_stats.math.easy	Easy math battles won	INT	Number	>=0	Y		
	battle_stats.math.average	Average math battles won	INT	Number	>=0	Y		
	battle_stats.math.difficult	Difficult math battles won	INT	Number	>=0	Y		
	battle_stats.science	science battle stats	Object	JSON	-	Y		
	battle_stats.science.easy	Easy science battles won	INT	Number	>=0	Y		
	battle_stats.science.average	Average science battles won	INT	Number	>=0	Y		
	battle_stats.science.difficult	Difficult science battles won	INT	Number	>=0	Y		
	challenge_stats	Challenge statistics object	Object	JSON	-	Y		
	challenge_stats.math	Math challenge stats	Object	JSON	-	Y		
	challenge_stats.math.easy	Easy math challenges completed	INT	Number	>=0	Y		

	challenge_stats.math.average	Average math challenges completed	INT	Number	>=0	Y		
	challenge_stats.math.difficult	Difficult math challenges completed	INT	Number	>=0	Y		
	challenge_stats.science	Science challenge stats	Object	JSON	-	Y		
	challenge_stats.science.easy	Easy science challenges completed	INT	Number	>=0	Y		
	challenge_stats.science.average	Average science challenges completed	INT	Number	>=0	Y		
	challenge_stats.science.difficult	Difficult science challenges completed	INT	Number	>=0	Y		
	memory_match_stats	General memory match stats	Object	JSON	-	Y		
	memory_match_stats.easy	Easy memory matches completed	INT	Number	>=0	Y		
	memory_match_stats.average	Average memory matches completed	INT	Number	>=0	Y		
	memory_match_stats.difficult	Difficult memory	INT	Number	>=0	Y		

		matches completed						
	puzzle_stats	Puzzle statistics object	Object	JSON	-	Y		
	puzzle_stats.solar_system	Solar system puzzle stats	Object	JSON	-	Y		
	puzzle_stats.solar_system.easy	Easy solar system puzzles completed	INT	Number	>=0	Y		
	puzzle_stats.solar_system.average	Average solar system puzzles completed	INT	Number	>=0	Y		
	puzzle_stats.solar_system.difficult	Difficult solar system puzzles completed	INT	Number	>=0	Y		
	created_at	Record Creation Stamp	DATE	ISO 8601	-	Y		
	updated_at	Record Last Update Stamp	DATE	ISO 8601	-	Y		
quiz_questions	_id	Unique ID	ObjectId	ObjectId	-	Y	PK	
	category	Question Category	String	Text	-	Y		
	difficulty_level	Difficulty Level	String	Text	easy/average/difficult	Y		
	question	Question	String	Text	-	Y		
	choice_a	Choice A	String	Text	-	Y		
	choice_b	Choice B	String	Text	-	Y		

	choice_c	Choice C	String	Text	-	Y		
	choice_d	Choice D	String	Text	-	Y		
	correct_answer	Correct Answer	String	Text	a/b/c/d	Y		
region	_id	Unique MongoDB ID	ObjectId	ObjectId	-	Y		
	id	Region ID	INT	Number	-	Y		
	region_name	Region Name	String	Text	-	Y		
province	_id	Unique MongoDB ID	ObjectId	ObjectId	-	Y		
	id	Province ID	INT	Number	-	Y		
	province_name	Province Name	String	Text	-	Y		
	region_id	Region ID	INT	Number	-	Y	FK	region
city	_id	Unique MongoDB ID	ObjectId	ObjectId	-	Y		
	id	City ID	INT	Number	-	Y		
	city_name	City Name	String	Text	-	Y		
	province_id	Province ID	INT	Number	-	Y	FK	province